Assignment-19.1

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Task 1 – Python to Java Conversion

Prompt:

#give me a simple python code for vote #voting system
Simple Voting System

- **CODE:**

```python
#TASK-1
#give me a simple python code for vote #voting system
# Simple Voting System
class VotingSystem:
    def __init__(self):
        self.votes = {}

    def cast_vote(self, candidate):
        if candidate in self.votes:
            self.votes[candidate] += 1
        else:
            self.votes[candidate] = 1

    def get_results(self):
        return self.votes
```

```java
import java.util.HashMap;
import java.util.Map;
public class VotingSystem {
    private Map<String, Integer> votes;

    public VotingSystem() {
        votes = new HashMap<>();
    }

    public void castVote(String candidate) {
        votes.put(candidate, votes.getOrDefault(candidate, 0) + 1);
    }

    public Map<String, Integer> getResults() {
        return votes;
    }
}
```

Explanation:
This task is about converting a Python voting system into Java.
The logic remains the same (store votes, count votes, show results), but the syntax and structure change according to Java (using classes, HashMap, and data types).

Task 2 – C++ to Python Function Conversion
Prompt:

//convert a c++ function for calculation into python
//ensure that the python version follows pythonic conventions and is efficient

CODE:

```cpp
#include <iostream>
#include <cmath>

double calculateDistance(double x1, double y1, double x2, double y2) {
    return std::sqrt(std::pow(x2 - x1, 2) + std::pow(y2 - y1, 2));
}
int main() {
    double x1, y1, x2, y2;
    std::cout << "Enter coordinates of the first point (x1 y1): ";
    std::cin >> x1 >> y1;
    std::cout << "Enter coordinates of the second point (x2 y2): ";
    std::cin >> x2 >> y2;

    double distance = calculateDistance(x1, y1, x2, y2);
    std::cout << "The distance between the two points is: " << distance << std::endl;

    return 0;
}
```

```python
import math
def calculate_distance(x1, y1, x2, y2):
    """Calculate the distance between two points (x1, y1) and (x2, y2)."""
    return math.sqrt((x2 - x1) ** 2 + (y2 - y1) ** 2)
# Example usage
if __name__ == "__main__":
    x1 = float(input("Enter coordinates of the first point (x1 y1): ").split()[0])
    y1 = float(input("Enter coordinates of the first point (x1 y1): ").split()[1])
    x2 = float(input("Enter coordinates of the second point (x2 y2): ").split()[0])
    y2 = float(input("Enter coordinates of the second point (x2 y2): ").split()[1])
    distance = calculate_distance(x1, y1, x2, y2)
    print(f"The distance between the two points is: {distance}")
```

Explanation:
This task is about converting a C++ function into Python code while keeping the same logic.
The program calculates the distance between two points using a mathematical formula, but rewritten using Python syntax and conventions.

Task 3 – SQL to Pandas Query

Prompt:
Generate a Python program using pandas to create a DataFrame with employee names and salaries.
Filter the data to display only employees with salary greater than 50000.
Print only the name and salary columns.

Code:

```python
import pandas as pd

data = {
    'name': ['John', 'Alice', 'Bob', 'Emma'],
    'salary': [45000, 60000, 75000, 40000]
}

df = pd.DataFrame(data)

result = df[df['salary'] > 50000][['name', 'salary']]

print(result.to_string(index=False))
```

```sql
SELECT name, salary
FROM employees
WHERE salary > 50000;
```

Explanation:
This task demonstrates how to filter data based on a condition using both Python (Pandas) and SQL.
In Python, we use a DataFrame to store data and apply conditions to filter rows, while in SQL, we use a select query with a where clause to achieve the same result.
Both approaches return employees whose salary is greater than 50,000, showing how similar operations can be performed in different technologies.

Task 4 – Real-Time Application: Multi-Language Snippet

Prompt :

Write a JavaScript program that fetches HTML content from a given URL using the fetch API.
Parse the HTML using DOMParser and extract all anchor (<a>) tag links.
Print all the extracted hyperlinks in the console.

CODE:

```python
import requests
from bs4 import BeautifulSoup

# URL of the website
url = "https://www.onlinegdb.com/"

# Send request to the website
response = requests.get(url)

# Parse HTML content
soup = BeautifulSoup(response.text, "html.parser")

# Extract all links
links = soup.find_all("a")

# Print all href links
for link in links:
    href = link.get("href")
    if href:  # check if link is not None
        print(href)
```

```javascript
fetch("https://www.onlinegdb.com/")
  .then(response => response.text())
  .then(data => {
    const parser = new DOMParser();
    const doc = parser.parseFromString(data, "text/html");

    const links = doc.querySelectorAll("a");

    links.forEach(link => {
      console.log(link.href);
    });
  })
  .catch(error => console.error("Error:", error));
```

Explanation: In this assignment, we used AI to convert code from one programming language to another. We converted Python to Java, C++ to Python, SQL to Pandas, and also worked on a real-time example.
We checked that all converted programs give correct output. This helped us understand differences between languages and how AI can make coding easier and faster.